



CSE3023 WEB-BASED APPLICATION DEVELOPMENT

PROGRAMME: COMPUTER SCIENCE (SOFTWARE ENGINEERING)

SEMESTER II 2025/2026

PROJECT TITLE:

SISWAPOST: SMART PARCEL MANAGEMENT SYSTEM

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1.0 INTRODUCTION

1.1 Project Background

SiswaPost: Smart Parcel Management System is a dynamic web-based application developed using the Java MVC (Model-View-Controller) architecture. The system is designed to improve and modernize parcel management processes in campus or residential environments.

The application utilizes Java Servlets as the Controller, JSP (JavaServer Pages) as the View, and structured Java Model classes coupled with Data Access Objects (DAO) to handle business logic and database communication. Additionally, Utility (Util) classes are integrated to handle input validation and helper tasks. This comprehensive structure ensures a strict separation of roles within the system, making the codebase significantly easier to maintain, test, and scale.

All parcel data is safely stored and managed in a relational database. This architecture enables real-time status updates, faster records searching, and highly improved data integrity while drastically reducing human operational errors. In short, SiswaPost transforms traditional campus parcel handling into a faster, highly accurate, and securely organized digital workflow.

1.2 Problem Statement

1.2.1 Inefficiency of Manual Tracking

Staff record parcel information manually in PDF files. This process takes time and is not efficient, especially when many parcels arrive at the same time. It is also difficult to search for specific parcel records quickly because the data is not stored in a structured system. As a result, staff may face delays when checking parcel details, and overall work becomes slower.

1.2.2 Lack of Automated Notification

Students are informed about parcel arrivals only through Telegram groups. This means students must always check the group manually to see if their parcel has arrived. Some students may miss the message or

do not use Telegram at all. This causes delays in parcel collection and creates inconvenience for students, as there is no direct or automatic notification system.

1.2.3 Operational Risk and Human Error

The current system uses both PDF records and paper forms during parcel collection. This manual process increases the chance of human error, such as incorrect data entry or missing information. Paper forms can also be lost or damaged and PDF files are not properly secured or backed up. Because of this, parcels may be misplaced or given to the wrong student, which affects trust and system reliability.

1.3 Project Objectives

- **To design and develop a structured web-based system** using the Java MVC architecture (Servlets, DAO, Model, and JSP) to replace slow, manual tracking methods and transition from unstructured files into a centralized relational database.
- **To implement an automated notification system** that instantly updates parcel statuses and alerts recipients, eliminating the reliance on manual group messaging and reducing delayed collections.
- **To minimize operational risks and human error** by deploying automated data validation mechanisms and secure digital CRUD workflows, replacing risky paper-based logs and unsecured files.
- **To improve administrative and user efficiency** through dedicated management modules that streamline parcel lifecycles, user registration, and notification tracking for both staff and students.

1.4 Scope of Projects

1.4.1 Target Users

- **Administrator (Admin)** : Counter staff or residential management personnel responsible for registering incoming parcels, updating collection statuses, and maintaining user records.

- **Students:** Campus residents who use the platform to track their arrived parcels, manage their user profiles, and receive status notifications.

1.4.2 Functional Scope

- **Parcel Management Module**
 - Allow administrator to perform full CRUD operations:
 - Register new incoming parcels (capturing tracking number, recipient matric number, and arrival date).
 - Update parcel lifecycle statuses (e.g., *Arrived*, *Pending Collection*, *Collected*, *Overdue*)
 - Remove duplicate or outdated records
 - Provides a searchable and filterable database view for students to track their specific parcel deliveries.
- **Notification Management Module**
 - Automatically generates and links notifications to a student's profile when an administrator updates a parcel status.
 - Allows students to view a personalized list of alerts, filter notifications by type, and manually toggle their status between *Read* and *Unread*.
 - Enables administrators to create, update, or purge old notifications to optimize system efficiency.
- **User Management Module**
 - Secures the system via a unified login and logout authentication workflow for both students and admins.
 - Enables new users to register accounts and empowers existing users to update profile fields like name, email, and password.
 - Grants administrators the authority to review active profiles and delete inactive accounts to ensure data accuracy

2.0 SYSTEM DESIGN

2.1 Sitemap / Navigate Structure

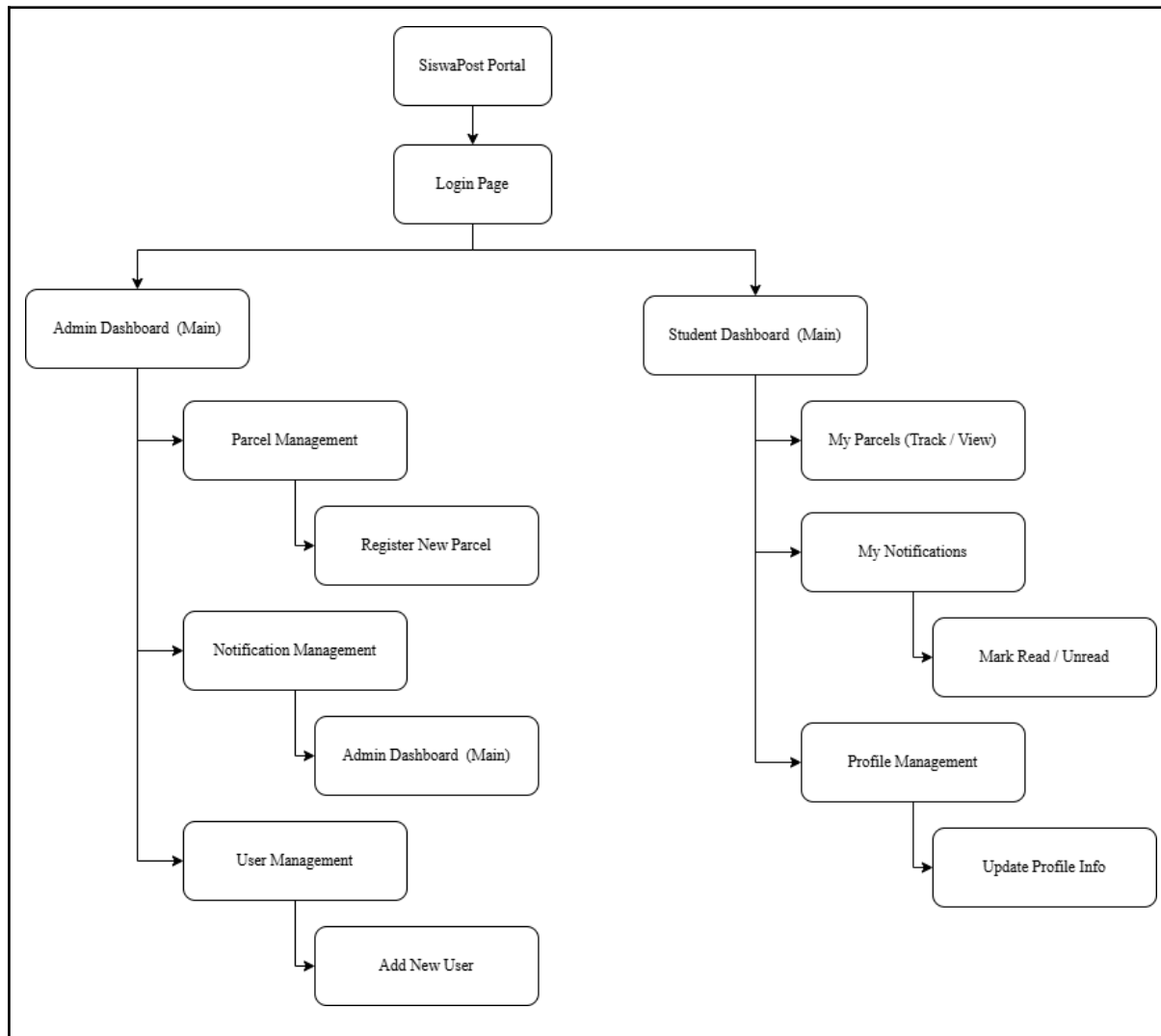


Figure 2.1 : Sitemap / Navigate Structure SiswaPost Portal

The Unified Authentication Gateway (Login Screen)

Every interaction with SiswaPost begins at the secure Login Page, where users are required to authenticate themselves before accessing the system. Students log in using their Matric Number while administrators use their Staff ID and password. The system verifies the submitted credentials against the database and upon successful authentication it creates a secure user session. Based on the user assigned role, the system automatically redirects them to either the Administrator Dashboard or the Student Dashboard, ensuring that each user can only access the features and information authorized for their account.

Administrator Navigation

After successfully logging in, administrators are directed to the Admin Dashboard, which serves as the main control centre of the system. A navigation sidebar provides quick and convenient access to the three main management modules:

- **Parcel Management**

This module allows administrators to manage parcel records throughout the delivery process. They can register newly arrived parcels, search for specific records, update parcel collection status, and remove duplicate or incorrect entries when necessary.

- **Notification Management**

This module enables administrators to manage notifications sent to students. Administrators can review notification history, resend notifications if required, and remove outdated notification records to keep the system organized.

- **User Management**

This module provides administrators with the ability to manage user accounts. They can add new users, update existing user information, and remove inactive accounts, ensuring that all registered users are properly maintained within the system.

Student Navigation

After successfully logging in, students are directed to the Student Dashboard, which is designed specifically for their role. Administrative functions are not accessible, ensuring the security and integrity of the system. The dashboard provides three main sections that allow students to manage their parcel-related information and personal account.

- **My Parcels**

This section displays all parcels associated with the logged-in student's matric number. Students can track the current status of their parcels and view relevant information. The page is read-only, meaning parcel records cannot be modified by students.

- **My Notifications**

This section serves as a personal notification centre where students receive updates about their parcels. Students can view notification details and manage their inbox by marking notifications as read or unread.

- **Profile Management**

This section allows students to manage their account information. They can update their personal details, change their contact email, and reset their password through a secure self-service process.

2.2 User Interface Design

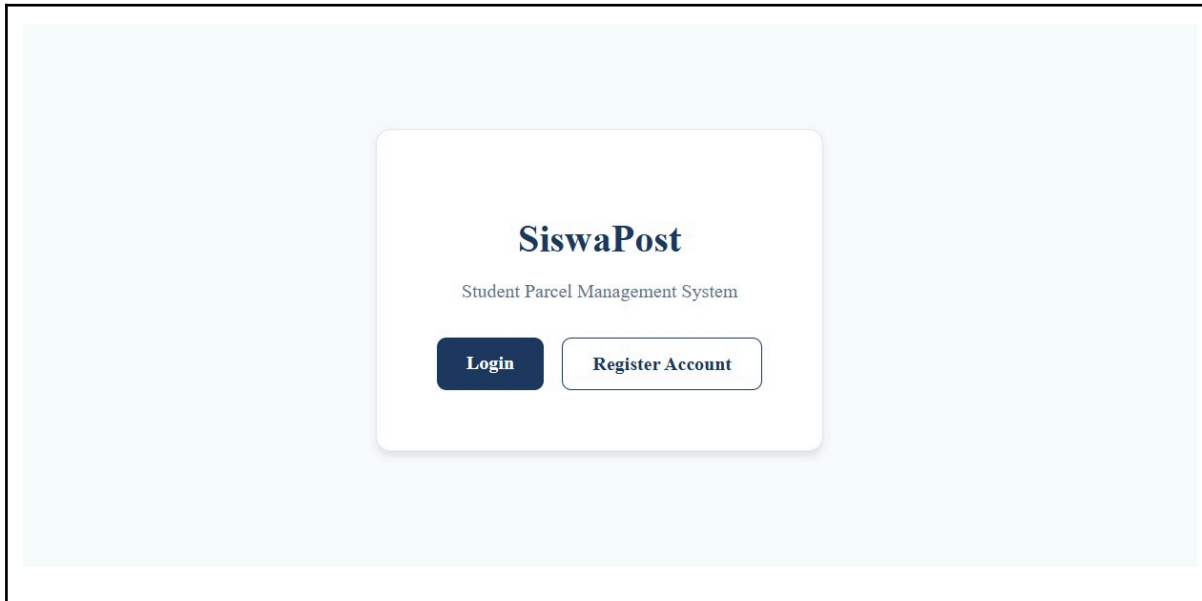
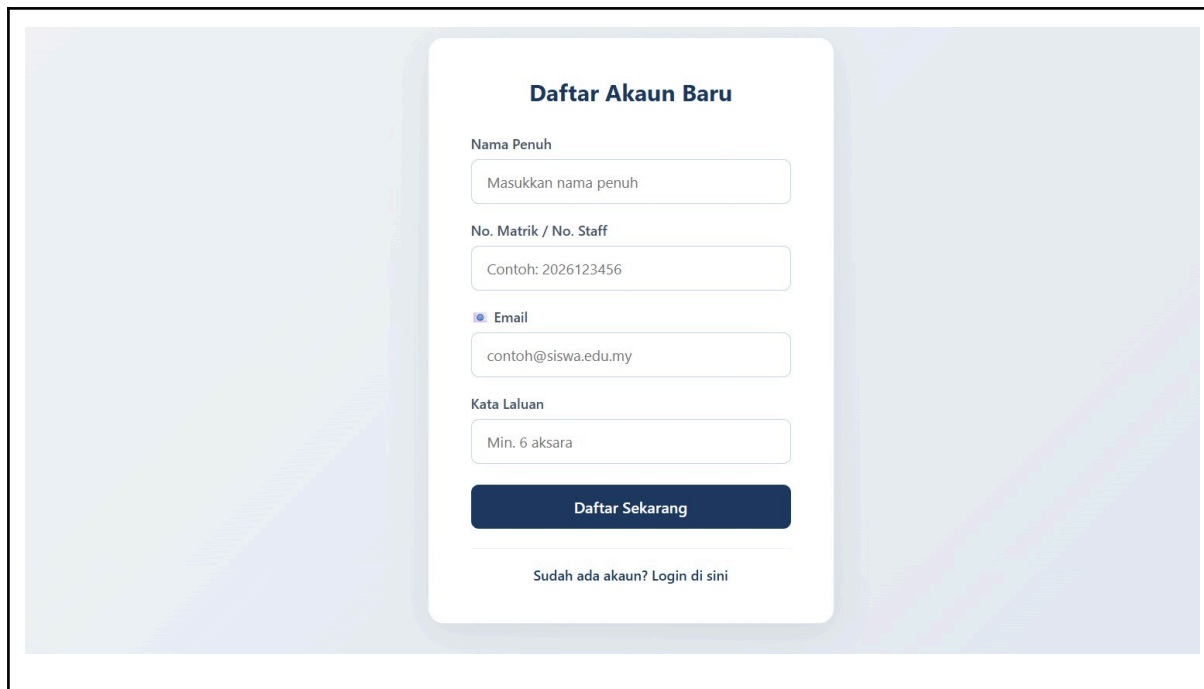


Figure 2.2.1 : Landing page SiswaPost system

This landing page serves as the main entry point to the SiswaPost system. It provides users with two primary options which is Login for existing students and administrators to access their respective dashboards and Register Account for new students to create an account.

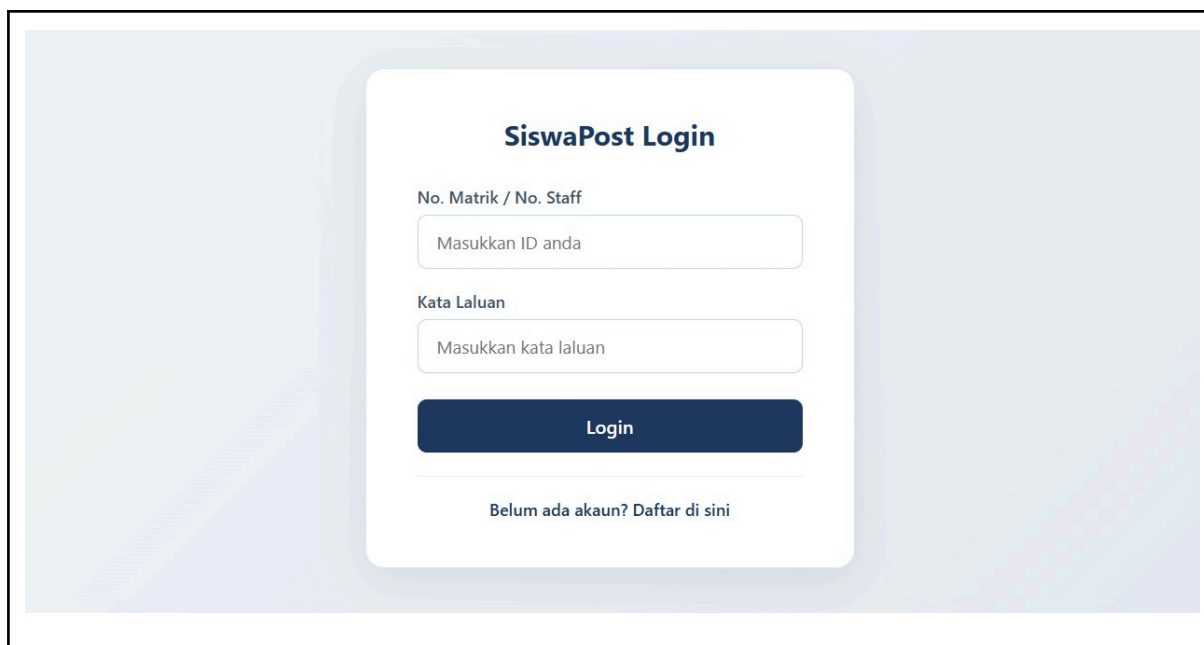


The image shows a registration form titled "Daftar Akaun Baru" (Register New Account). It is a white card with rounded corners on a light blue background. The form contains the following fields and elements:

- Nama Penuh**: A text input field with the placeholder "Masukkan nama penuh".
- No. Matrik / No. Staff**: A text input field with the placeholder "Contoh: 2026123456".
- Email**: A text input field with the placeholder "contoh@siswa.edu.my". A small blue circle icon is to the left of the label.
- Kata Laluan**: A text input field with the placeholder "Min. 6 aksara".
- Daftar Sekarang**: A dark blue button with white text.
- Sudah ada akaun? Login di sini**: A link at the bottom of the form.

Figure 2.2.2: Register Account page

The Daftar Akaun Baru (Register New Account) page enables new users to register by providing their personal information. Existing users can return to the login page using the provided link.



The image shows a login form titled "SiswaPost Login". It is a white card with rounded corners on a light blue background. The form contains the following fields and elements:

- No. Matrik / No. Staff**: A text input field with the placeholder "Masukkan ID anda".
- Kata Laluan**: A text input field with the placeholder "Masukkan kata laluan".
- Login**: A dark blue button with white text.
- Belum ada akaun? Daftar di sini**: A link at the bottom of the form.

Figure 2.2.3 : Login Account page

The SiswaPost Login page enables existing students and administrators to securely access their accounts by entering their credentials, while providing a link for new users to register.

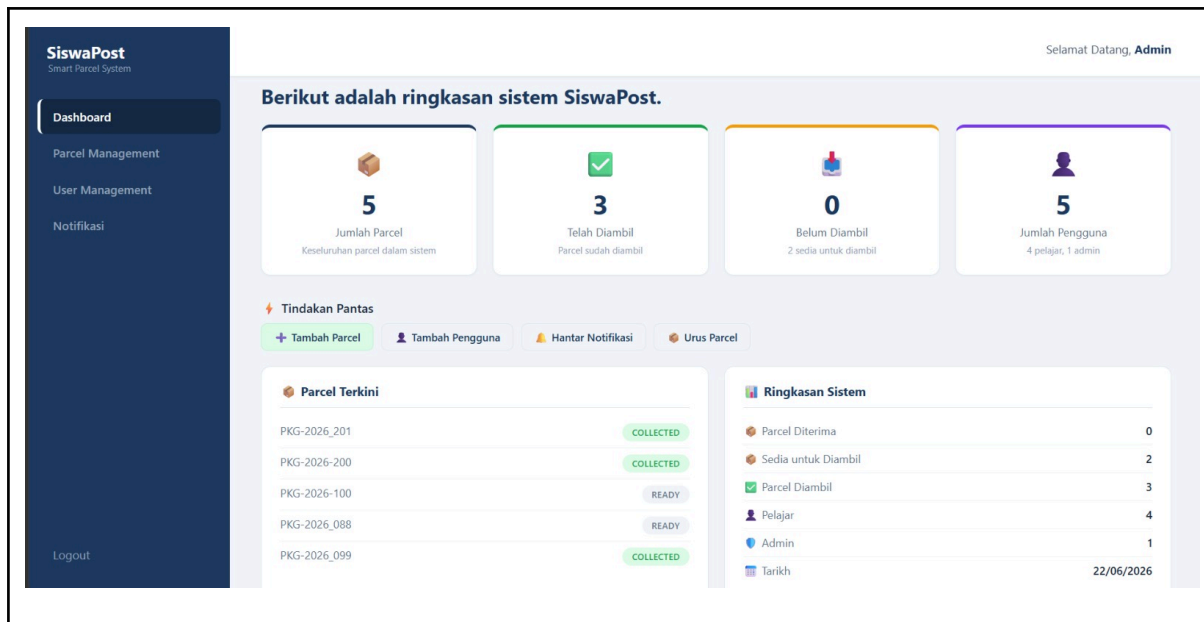


Figure 2.2.4 : Dashboard page (Admin)

The Admin Dashboard page provides administrators with a centralized summary of parcel metrics and quick action buttons to streamline daily campus management workflows.

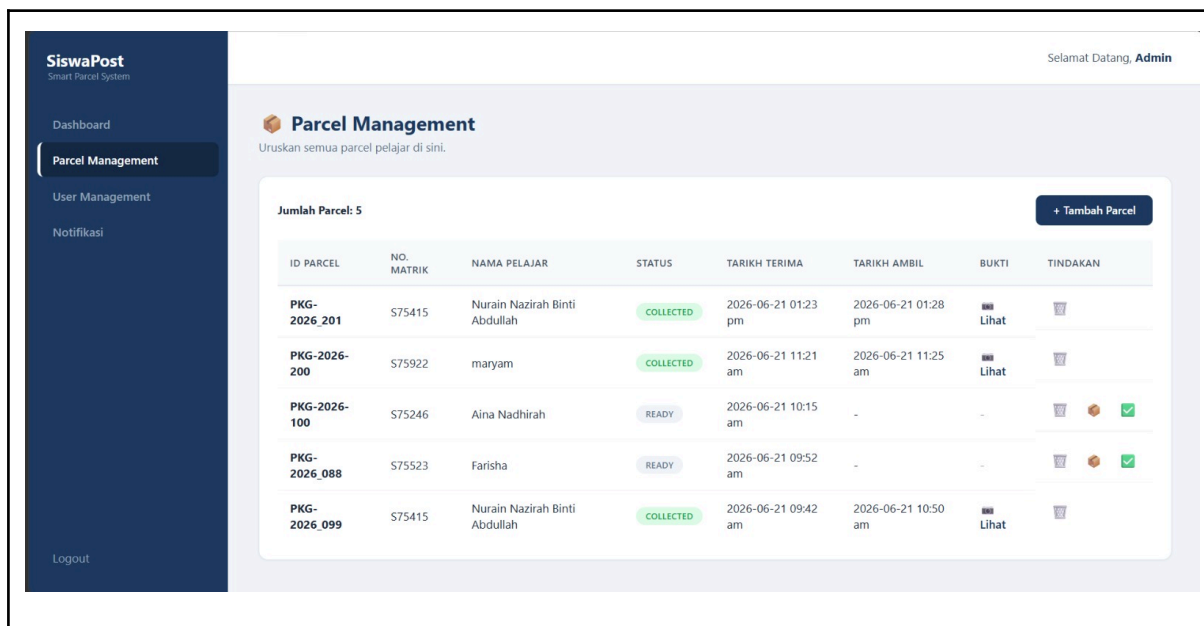


Figure 2.2.5 : Parcel Management page (Admin)

The Admin Parcel Management page allows administrators to perform full CRUD operations by registering incoming packages, updating delivery status badges, searching records, and deleting duplicate entries.

+ Tambah Parcel Baru

ID Parcel

Contoh: PKG-2026-001

No. Matrik Pelajar

Masukkan no. matrik

Pastikan no. matrik wujud dalam sistem

Nama Pelajar

Masukkan nama pelajar

Status

Received

Batal Tambah

Figure 2.2.6 : Register Parcel (Admin)

The Tambah Parcel page enables administrators to register a new incoming package by entering the tracking number and recipient's matric number, automatically recording the arrival date into the database.

SiswaPost
Smart Parcel System

Selamat Datang, **Admin**

Dashboard
Parcel Management
User Management
Notifikasi

User Management
Uruskan semua pengguna sistem di sini.

Jumlah Pengguna: 5 [+ Tambah Pengguna](#)

ID	NAMA PENUH	NO. MATRIK	EMAIL	ROLE	STATUS	TINDAKAN
1	Admin	ADMIN001	admin@siswapost.com	admin	ACTIVE	
5	Ainaa Nadhirah	S75246	aina@gmail.com	student	ACTIVE	
4	Farisha	S75523	pica@gmail.com	student	ACTIVE	
6	maryam	S75922	yam@gmail.com	student	ACTIVE	
3	Nurain Nazirah Binti Abdullah	S75415	ain@gmail.com	student	ACTIVE	

Logout

Figure 2.2.7 : User Management page (Admin)

The Admin User Management page provides administrators with a structured directory to monitor registered profiles, update account metadata or remove inactive credentials securely.

SiswaPost
Smart Parcel System

Selamat Datang, **Admin**

Sistem Notifikasi (Admin)

Berikut adalah senarai notifikasi terkini anda.

Jumlah: 6 + Tambah Notifikasi

TARIKH	NO. MATRIK	JENIS	MESEJ	STATUS	TINDAKAN
2026-06-21 11:24 am	S75922	parcel_ready	mri amik	READ	
2026-06-21 10:42 am	S75415	parcel_collected	dah collect	READ	
2026-06-21 10:41 am	S75415	parcel_ready	mai amik	READ	
2026-06-21 10:16 am	S75246	parcel_collected	dah boleh amikk barangg	READ	
2026-06-21 09:55 am	S75523	parcel_arrived	Barang dh sampai	READ	
2026-06-21 09:42 am	S75415	parcel_arrived	Parcel Arrived	READ	

Logout

Figure 2.2.8 : Notification page (Admin)

The Admin Notification page enables administrators to track system logs, view alert transaction histories, create new notification alerts, and delete historical communication records.

Tambah Notifikasi Baru

No. Matrik

ID Parcel (Optional)

Jenis Notifikasi

General

Mesej

Figure 2.2.9 : Add New Notification page (Admin)

The Tambah Notifikasi Baru page allows administrators to dispatch a custom alert to a specific resident by providing the student's matric number, notification type and message content.

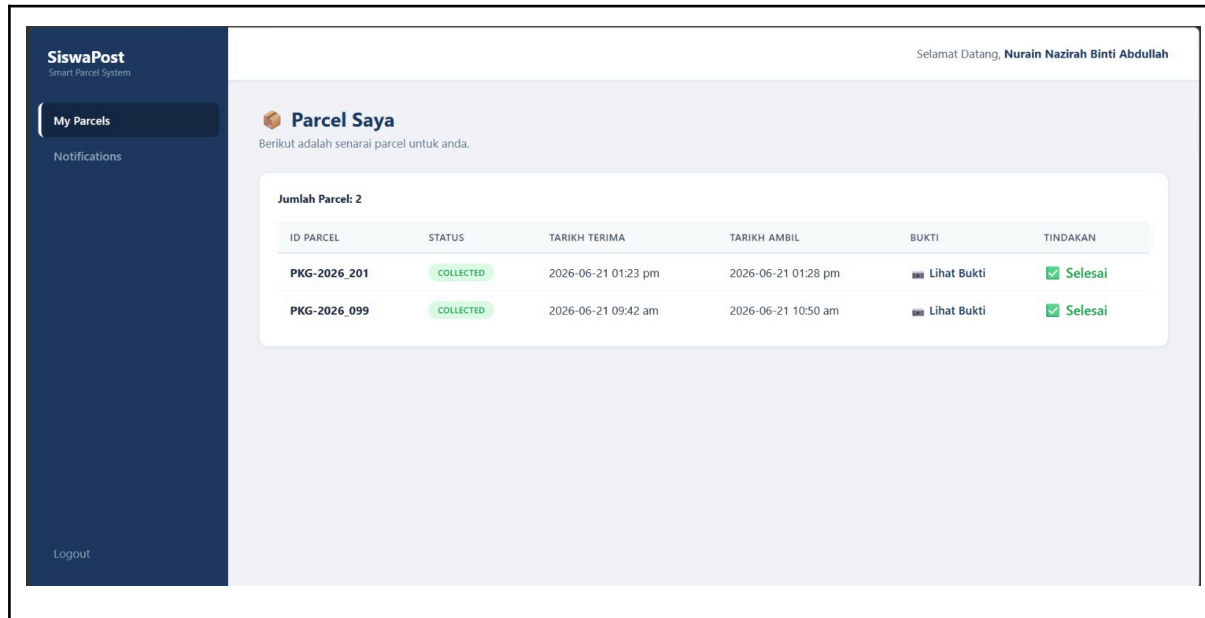


Figure 2.2.10 : My Parcel page (Student)

The Parcel Saya page provides students with a personalized tracking view to monitor their own arrival records, collection timestamps, status updates and uploaded pickup confirmation receipts.

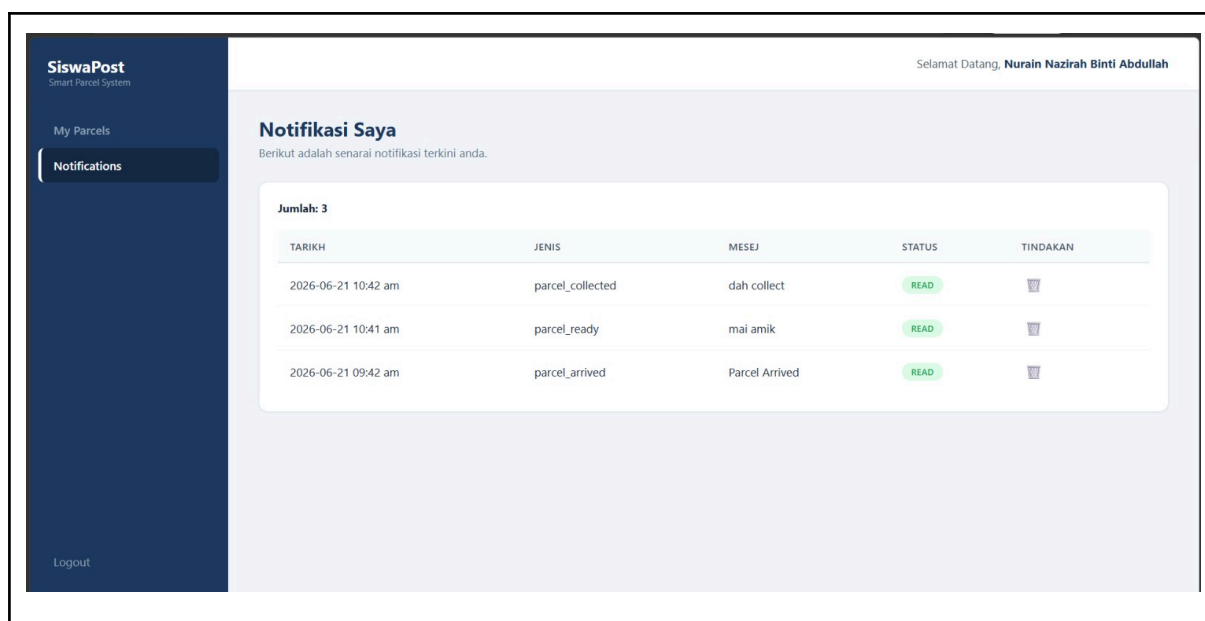


Figure 2.2.11 : Notification page (Student)

The Notifikasi Saya page enables students to view automated, read-only system status alerts directed to them, track message types, and delete read alert records from their personal inbox grid.

2.3 Description of Major Pages and Functionalities

2.3.1 Core Gateway Pages

- **Core Landing Portal:** The SiswaPost Landing page acts as an entry router providing direct buttons for existing users to log in or for new student residents to create an account.
- **SiswaPost Login Page:** The SiswaPost Login page enables existing students and administrators to securely access their accounts by entering their credentials, while providing a link for new users to register.
- **Daftar Akaun Baru (Register Account) Page:** The Daftar Akaun Baru page enables new users to register by providing their personal information, while existing users can return to the login page using the provided link.

2.3.2 Administrator Management Modules

- **Admin Dashboard Page:** The Admin Dashboard page provides administrators with a centralized summary of parcel metrics and quick action buttons to streamline daily campus management workflows.
- **Admin Parcel Management Page:** The Admin Parcel Management page allows administrators to perform full CRUD operations by registering incoming packages, updating delivery status badges, searching records, and deleting duplicate entries.
- **Tambah Parcel Baru (Add New Parcel) Modal:** The Tambah Parcel page enables administrators to register a new incoming package by entering the tracking number and recipient's matric number, automatically recording the arrival date into the database.
- **Admin User Management Page:** The Admin User Management page provides administrators with a structured directory to monitor registered profiles, update account metadata, or remove inactive credentials securely.

- **Admin Notification Page:** The Admin Notification page enables administrators to track system logs, view alert transaction histories, create new notification alerts, and delete historical communication records.
- **Tambah Notifikasi Baru (Add New Notification) Modal:** The Tambah Notifikasi Baru page allows administrators to dispatch a custom alert to a specific resident by providing the student's matric number, notification type, and message content.

2.3.3 Student Self-Service Modules

- **Parcel Saya (My Parcels) Page:** The Parcel Saya page provides students with a personalized tracking view to monitor their own arrival records, collection timestamps, status updates, and uploaded pickup confirmation receipts.
- **Notifikasi Saya (My Notifications) Page:** The Notifikasi Saya page enables students to view automated, read-only system status alerts directed to them, track message types and delete read alert records from their personal inbox grid.

3.0 SYSTEM IMPLEMENTATION

3.1 Development tools used

In order to have a systematic and isolated process of development, the below mentioned integrated tools were used during the implementation phase of SiswaPost:

- **Integrated Development Environment (IDE):** NetBeans IDE was used as the main development environment where it provides built-in facilities such as project structuring, Java components refactorings, and direct deployment of web archives in the local servlet containers.
- **Java Development Kit (JDK):** JDK 21 was configured for compiling source codes as it provides better runtime performance and more security primitives than previous versions.
- **Web Server & Servlet Container:** Apache Tomcat was configured as a local web application server for hosting the servlet contexts and dynamically converting the JavaServer Pages (JSP) into Java servlets.

- Database Management System (DBMS): MySQL was used as the main repository for storing data in a centralized way.
- Database Administration Interface: phpMyAdmin was used for designing database schema, defining primary and foreign keys, and testing SQL queries.

3.2 Technologies used

SiswaPost system was developed utilizing a decoupled production-grade technology stack in accordance with the typical enterprise Java MVC (Model-View-Controller) framework:

- Backend/Controller Layer: The Java Servlets processes HTTP requests, performs session-based authentication, invokes back-end application logic, and determines routing of the user view.
- Data Access & Enterprise Logic (Model Layer):
 1. **Java Model Classes:** Structured Java classes that represent system entities (e.g., User, Parcel, Notification) encapsulating properties using variables along with standard getter and setter methods to hold data.
 2. **Data Access Objects (DAO):** Implement structured SQL statements using JDBC (Java Database Connectivity) to decouple basic database transactions (CRUD) from servlet routes.
 3. **Utility (Util) Classes:** Integrated to handle foundational helper tasks such as input validation and date-time formatting.
- Frontend Layer - Java Server Pages (JSP): Provides dynamic presentation layer rendering the database statistics in a secure way based on parameters provided by controllers.
 1. **JavaServer Pages Standard Tag Library (JSTL):** Used in JSP files to safely iterate over back-end data collections using the tags library without utilizing scriptlets (<% %>) in JSP file.
 2. **CSS and JavaScript:** Used to provide a modern, user-friendly and responsive administration sidebar and student tracking layout interface.

3.3 Description of implemented features

The system fulfills the complete programmatic functional scope across three decoupled self-service modules:

Parcel Management Module:

- Allows the Admin to perform full CRUD operations: registering incoming parcels (tracking number, recipient matric number, arrival date), updating lifecycle statuses (*Arrived*, *Pending Collection*, *Collected*, *Overdue*), and removing duplicate entries.
- Provides a searchable and filterable database view for students to track their specific parcel delivery statuses in real time.

Notification Management Module:

- Automatically generates and links notification alerts to a student's profile whenever an administrator updates a parcel status.
- Allows students to view a personalized list of alerts, filter notifications, and manually toggle their status between *Read* and *Unread*.
- Enables administrators to create, update, or purge old notification history records.

User Management Module:

- Secures the system via a unified login and logout authentication workflow for both students and admins.
- Enables new student residents to register accounts and empowers existing users to update profile fields like name, email, and password.
- Grants administrators the authority to review active profiles and delete inactive accounts to ensure data accuracy.

4.0 RESULTS

4.1 Results obtained

The SiswaPost: Smart Parcel Management System was successfully developed based on the proposed design and requirements. The system is able to manage parcel records efficiently through a structured database using Java MVC architecture, including Servlets, JSP, Model classes, and DAO layers.

The Parcel Management Module allows administrators to register new parcels, update parcel statuses, and manage existing records effectively. Students are able to view their parcel information in real time, ensuring faster and more accurate tracking compared to the previous manual method.

The Notification Management Module successfully generates updates when parcel statuses are changed, allowing students to receive timely information regarding their parcel arrival and collection status.

In addition, the User Management Module ensures proper authentication and profile handling for both administrators and students, improving system security and data integrity.

4.2 Discussion on whether project objectives achieved

All project objectives have been successfully achieved. The system effectively replaces the manual parcel tracking process with a centralized web-based platform, improving efficiency and reducing human error.

The implementation of automated notifications has improved communication between administrators and students, reducing delays in parcel collection. Furthermore, the use of structured CRUD operations and database integration has minimized risks related to data inconsistency and loss.

Overall, the system meets its intended goals in improving parcel management efficiency, accuracy, and accessibility within the campus environment.

5.0 CONCLUSION

5.1 Summary of the project

SiswaPost: Smart Parcel Management System is a web-based application developed using Java MVC architecture to replace traditional manual parcel handling methods. The system integrates Servlets as Controller, JSP as View, and Model-DAO structure for database interaction, ensuring a clean separation of concerns and better system maintainability.

Overall, the system successfully digitizes parcel management processes by centralizing data storage, improving record accuracy, and enabling real-time updates for parcel status. With the addition of automated notification features, both administrators and students can manage and track parcel information more efficiently within a single platform.

5.2 Strength and limitations

Strengths:

The system provides a structured and organized parcel management workflow, reducing dependency on manual documentation such as PDFs and paper forms. It improves data integrity through relational database usage and minimizes human errors via controlled CRUD operations and validation logic. The notification module also enhances communication efficiency by ensuring students receive timely updates without relying on external messaging platforms.

Limitations:

The system still depends on manual input from administrators when registering parcels, which may introduce delays if data entry is not performed consistently. In addition, the notification system is currently limited to in-system alerts and does not yet support external integrations such as email or mobile push notifications. The system is also designed for a specific campus environment, meaning scalability to larger institutions may require further optimization.

5.3 Future improvement

Future enhancements for SiswaPost could include integration with automated scanning technologies such as QR code or barcode systems to speed up parcel registration and reduce manual input. Expanding the notification system to include email and mobile push notifications would improve accessibility and ensure students receive updates instantly.

In addition, the system can be enhanced by implementing role-based access control (RBAC) with more granular permissions to improve system security and ensure only authorized users can perform sensitive operations.

The system could also be upgraded into a cloud-based architecture to improve scalability, data backup reliability, and multi-campus support in the future.